

Part A: Complete each question, showing work in the space provided. Do not attach additional pages. Each question in this part is worth 5 pts.

1. Use back-substitution to solve the system of linear equations. Write your solution as an ordered triple.

$$-x + y - z = 0$$

$$7y + z = 5$$

$$\frac{1}{7}z = 0$$

Step 1: $\frac{1}{7}z = 0 \rightarrow z = 0$

Step 2: $7y + (0) = 5 \rightarrow 7y = 5 \rightarrow y = \frac{5}{7}$

Step 3: $-x + \left(\frac{5}{7}\right) - (0) = 0 \rightarrow -x = -\frac{5}{7}$
 $x = \frac{5}{7}$

$$(x, y, z) = \left(\frac{5}{7}, \frac{5}{7}, 0\right)$$

2. Explain why the given system of equations must have at least one solution (without solving the system).

$$3x + 3y - z = 0$$

$$6x + 3y + 6z = 0$$

$$3x + 9y - 17z = 0$$

Because the system is homogeneous, it is guaranteed to have the trivial solution $(0, 0, 0)$, which satisfies all equations. This means the system must be consistent and has at least one solution.

Part B: Complete each question, showing work in the space provided. Do not attach additional pages. Each question in this part is worth 10 pts.

3. Use an inverse matrix to solve the system of linear equations. Write your solution as an ordered pair.

$$x_1 + 5x_2 = -2$$

$$-5x_1 - 4x_2 = -11$$

$$A = \begin{bmatrix} 1 & 5 \\ -5 & -4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ -11 \end{bmatrix}$$

For a 2×2 matrix, if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

$$\text{So, } A^{-1} = \frac{1}{(1)(-4) - (5)(-5)} \begin{bmatrix} -4 & -5 \\ 5 & 1 \end{bmatrix} = \frac{1}{21} \begin{bmatrix} -4 & -5 \\ 5 & 1 \end{bmatrix}$$

Now, compute $x = A^{-1}b$: $x = \frac{1}{21} \begin{bmatrix} -4 & -5 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -11 \end{bmatrix}$

1st entry: $(-4)(-2) + (-5)(-11) = 8 + 55 = 63$

2nd entry: $(5)(-2) + (1)(-11) = -10 - 11 = -21$

$$\text{So, } x = \frac{1}{21} \begin{bmatrix} 63 \\ -21 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$(x_1, x_2) = (3, -1)$$

4. Solve the system of equations using either Gaussian elimination with back-substitution or Gauss-Jordan elimination. If there are no solutions, write No Solution. If there are an infinite number of solutions, express x_1 , x_2 , and x_3 in terms of the parameter t .

$$\begin{aligned} x_1 + x_2 - 5x_3 &= 3 \\ x_1 - 2x_3 &= 1 \\ 2x_1 - x_2 - x_3 &= 0 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -5 & 3 \\ 1 & 0 & -2 & 1 \\ 2 & -1 & -1 & 0 \end{array} \right]$$

Step 1: $-R_1 + R_2 \rightarrow R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & -5 & 3 \\ 0 & -1 & 3 & -2 \\ 2 & -1 & -1 & 0 \end{array} \right]$$

Step 5: $3R_2 + R_3 \rightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 6: This corresponds to,

$$\begin{aligned} x_1 - 2x_3 &= 1 \\ x_2 - 3x_3 &= 2 \end{aligned}$$

let the free variable be, $x_3 = t$

Then, $x_1 = 1 + 2t$

$x_2 = 2 + 3t$

$x_3 = t$

$$(x_1, x_2, x_3) = (1 + 2t, 2 + 3t, t)$$

5. Find the inverse of the matrix.

$$A = \begin{bmatrix} 5 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

Use Gauss-Jordan on $[A|I]$ to find A^{-1} .

$$\left[\begin{array}{ccc|ccc} 5 & 2 & 4 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 4 & 2 & 5 & 0 & 0 & 1 \end{array} \right]$$

1.) $R_1 \leftrightarrow R_2$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 0 & 1 & 0 \\ 5 & 2 & 4 & 1 & 0 & 0 \\ 4 & 2 & 5 & 0 & 0 & 1 \end{array} \right]$$

3.) $2R_3 + R_2 \rightarrow R_2$
 $-2R_3 + R_1 \rightarrow R_1$
 $-R_2 \rightarrow R_2$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 0 & 5 & -2 \\ 0 & 1 & 0 & -2 & 9 & -2 \\ 0 & 0 & 1 & 0 & -2 & 1 \end{array} \right]$$

4.) $-R_2 + R_1 \rightarrow R_1$
 $\frac{1}{2}R_1 \rightarrow R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & 0 \\ 0 & 1 & 0 & -2 & 9 & -2 \\ 0 & 0 & 1 & 0 & -2 & 1 \end{array} \right]$$

Now, the left side of the matrix is I and the right side is A^{-1} .

$$A^{-1} = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 9 & -2 \\ 0 & -2 & 1 \end{bmatrix}$$

2.) $-5R_1 + 2R_2 \rightarrow R_2$
 $-2R_1 + R_3 \rightarrow R_3$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 2 & 0 & 1 & 0 \\ 0 & -1 & -2 & 2 & -5 & 0 \\ 0 & 0 & 1 & 0 & -2 & 1 \end{array} \right]$$

Part C: Application Questions (10 pts each). Choose 6 to solve. You may choose to solve an additional 1 question for up to 5 points of extra credit on this exam. Cross off the questions you don't want considered in your grade. Circle the question you want considered for extra credit.

For questions in part C, I do NOT need to see row operations. Show the equations when applicable, the initial matrix, and the row-reduced matrix.

1. Two planes start from Los Angeles International Airport and fly in opposite directions. The second plane starts $\frac{1}{2}$ hour after the first plane, but its speed is 20 kilometers per hour faster. Two hours after the first plane departs, the planes are 3110 kilometers apart. Find the airspeed of each plane.

Let the first plane's speed be V in km/h

Let the second plane's speed be $V+20$ in km/h.

$$2.) \text{ Solve: } 2V + \frac{3}{2}V + 30 = 3110$$

$$\frac{7}{2}V = 3080$$

$$V = 880$$

1.) plane 1 has flown 2 hours, distance of $2V$

Plane 2 has flown $2 - \frac{1}{2} = \frac{3}{2}$ hours, distance of $\frac{3}{2}(V+20)$.

$$\text{Thus, } 2V + \frac{3}{2}(V+20) = 3110$$

So: first plane: $V = 880$ km/h

Second plane: $V+20 = 900$ km/h

2. Determine the polynomial of least degree whose graph passes through the points $(-1, 2)$, $(0, 0)$, $(1, 1)$, $(4, 58)$.

To pass through 4 points, the least possible degree is at most 3, so $p(x) = ax^3 + bx^2 + cx + d$.

Since the graph passes through $(0, 0)$, $p(0) = d = 0$.

$$\text{So: } p(x) = ax^3 + bx^2 + cx.$$

Set up the equations: using $(-1, 2) \rightarrow -a + b - c = 2$

$$\text{using } (1, 1) \rightarrow a + b + c = 1$$

$$\text{using } (4, 58) \rightarrow 64a + 16b + 4c = 58$$

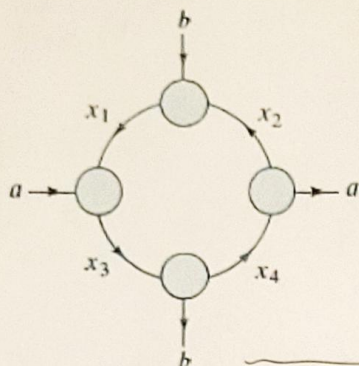
$$\text{Initial matrix: } \left[\begin{array}{ccc|c} -1 & 1 & -1 & 2 \\ 1 & 1 & 1 & 1 \\ 64 & 16 & 4 & 58 \end{array} \right]$$

$$\text{So: } a = \frac{3}{5}, b = \frac{3}{2}, c = -\frac{11}{10}, d = 0$$

$$\text{RREF Matrix: } \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3/5 \\ 0 & 1 & 0 & 3/2 \\ 0 & 0 & 1 & -11/10 \end{array} \right]$$

$$\text{Final polynomial: } p(x) = \frac{3}{5}x^3 + \frac{3}{2}x^2 - \frac{11}{10}x$$

3. The figure shows the flow of traffic (in vehicles per hour) through a network of street. Assume $a = 300$ and $b = 600$. Solve this system for the variables x_1, x_2, x_3, x_4 . If the system has an infinite number of solutions, express the variables in terms of the parameter t . Then find the traffic flow when $x_1 = 2x_2$.



Initial augmented matrix:
$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -600 \\ 0 & 1 & 0 & -1 & 300 \\ 0 & 0 & -1 & 1 & 600 \\ -1 & 0 & 1 & 0 & -300 \end{array} \right]$$

RREF:
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -300 \\ 0 & 1 & 0 & -1 & 300 \\ 0 & 0 & 1 & -1 & -600 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Thus x_4 is free.

Let $x_4 = t$. Then,

$$x_1 = t - 300, x_2 = t + 300, x_3 = t - 600, x_4 = t$$

$$(x_1, x_2, x_3, x_4) = (t - 300, t + 300, t - 600, t)$$

When $x_1 = 2x_2$, substitute into the parameter forms:

$$t - 300 = 2(t + 300) \rightarrow t = -900$$

Plus back in:

$$\text{Thus, } x_1 = -1200, x_2 = -600, x_3 = -1500, x_4 = -900$$

4. Use a system of equations to find the partial fraction decomposition of the rational expression. Then solve the system using matrices.

$$\frac{4x^2 + 3x - 3}{(x-1)(x+1)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

Step 1: Clear denominators

$$4x^2 + 3x - 3 = A(x^2 + 2x + 1) + B(x-1)(x+1) + C(x-1)$$

Expand each piece:

$$A(x+1)^2 = Ax^2 + 2Ax + A$$

$$B(x-1)(x+1) = Bx^2 - B$$

$$C(x-1) = Cx - C$$

Add them:

$$4x^2 + 3x - 3 = (A+B)x^2 + (2A+C)x + (A-B-C)$$

Step 2: Match coefficients to form a system.

$$\begin{cases} A+B=4 \\ 2A+C=3 \\ A-B-C=-3 \end{cases}$$

Step 3: Solve using matrices.

initial:
$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 2 & 0 & 1 & 3 \\ 1 & -1 & -1 & -3 \end{array} \right] \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ -3 \end{bmatrix}$$

RREF:
$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$$
 so, $A=1, B=3, C=1$

Final partial fraction decomposition:

$$\frac{4x^2 + 3x - 3}{(x-1)(x+1)^2} = \frac{1}{x-1} + \frac{3}{x+1} + \frac{1}{(x+1)^2}$$

5. The market research department at a manufacturing plant determines that 30% of the people who purchase the plant's product during any month will not purchase it the next month. On the other hand, 40% of the people who do not purchase the product during any month will purchase it the next month. In a population of 1000 people, 100 people purchased the product this month. How many will purchase the product next month and in 2 months?

Let P = purchases and N = does not purchase

1.) $P \rightarrow N = 0.30$, $P \rightarrow P = 0.70$

$N \rightarrow P = 0.40$, $N \rightarrow N = 0.60$

2.) Transition matrix: $T = \begin{bmatrix} 0.70 & 0.40 \\ 0.30 & 0.60 \end{bmatrix}$

initial state vector: $x_0 = \begin{bmatrix} 100 \\ 900 \end{bmatrix}$

3.) Next month: $x_1 = T x_0 = \begin{bmatrix} 0.70 & 0.40 \\ 0.30 & 0.60 \end{bmatrix} \begin{bmatrix} 100 \\ 900 \end{bmatrix} = \begin{bmatrix} 430 \\ 570 \end{bmatrix}$

So, 430 people will purchase next month.

4.) In 2 months: $x_2 = T x_1 = \begin{bmatrix} 0.70 & 0.40 \\ 0.30 & 0.60 \end{bmatrix} \begin{bmatrix} 430 \\ 570 \end{bmatrix} = \begin{bmatrix} 529 \\ 471 \end{bmatrix}$

So, 529 people will purchase in 2 months

6. In a population of 400,000 people, 120,000 are infected with a virus. After a person becomes infected and then recovers, the person is immune (cannot become infected again). Of the people who are infected, 4% will die each year and the others will recover. Of the people who have never been infected, 30% will become infected each year. How many people will be infected in 4 years? Show how you set up this problem, but feel free to use technology for the matrix multiplication. (Round your answer to the nearest whole number.)

Let S_n = susceptible

I_n = infected

R_n = recovered and immune

D_n = deceased

Thus, $S_n + I_n + R_n + D_n = 400,000$

1.) $I \rightarrow D = 0.04$, $I \rightarrow R = 0.96$, $I \rightarrow I = 0$

$S \rightarrow I = 0.30$, $S \rightarrow S = 0.70$

Transition matrix: $T = \begin{bmatrix} 0.70 & 0 & 0 & 0 \\ 0.30 & 0 & 0 & 0 \\ 0 & 0.96 & 1 & 0 \\ 0 & 0.04 & 0 & 1 \end{bmatrix}$

3.) $x_1 = T x_0 = \begin{bmatrix} 196000 \\ 84000 \\ 115200 \\ 4800 \end{bmatrix}$

$x_3 = T^3 x_0 = \begin{bmatrix} 96040 \\ 41160 \\ 252288 \\ 10512 \end{bmatrix}$

$x_2 = T^2 x_0 = \begin{bmatrix} 137200 \\ 58800 \\ 195840 \\ 8160 \end{bmatrix}$

$x_4 = T^4 x_0 = \begin{bmatrix} 67228 \\ 28812 \\ 291801.6 \\ 12158.4 \end{bmatrix}$

Thus, the number of infected in 4 years is $I_4 = 28812$ people.

2.) Initial vector $x_0 = \begin{bmatrix} S_0 \\ I_0 \\ R_0 \\ D_0 \end{bmatrix} = \begin{bmatrix} 280,000 \\ 120,000 \\ 0 \\ 0 \end{bmatrix}$

7. A system composed of two industries, coal and steel, has the following input requirements.

- (a) To produce \$1.00 worth of output, the coal industry requires \$0.20 of its own product and \$0.40 of steel.
 (b) To produce \$1.00 worth of output, the steel industry requires \$0.30 of its own product and \$0.60 of coal.

Find D , the input-output matrix for the system. Then solve for the output matrix X in the equation $x = DX + E$, where E is the external demand matrix $E = \begin{bmatrix} 10,000 \\ 20,000 \end{bmatrix}$. Round to the nearest whole number.

Step 1: Write in Leontief form; $x = Ax + E$
 $(I - A)x = E$

$$A = \begin{bmatrix} 0.20 & 0.60 \\ 0.40 & 0.30 \end{bmatrix} \quad E = \begin{bmatrix} 10000 \\ 20000 \end{bmatrix}$$

Step 2: form matrix $I - A$:

$$I - A = \begin{bmatrix} 1 - 0.20 & -0.60 \\ -0.40 & 1 - 0.30 \end{bmatrix} = \begin{bmatrix} 0.80 & -0.60 \\ -0.40 & 0.70 \end{bmatrix}$$

Step 3: solve for $x = (I - A)^{-1} E$

$$(I - A)^{-1} = \frac{1}{0.32} \begin{bmatrix} 0.70 & 0.60 \\ 0.40 & 0.80 \end{bmatrix} = 3.125 \begin{bmatrix} 0.70 & 0.60 \\ 0.40 & 0.80 \end{bmatrix}$$

$$x = 3.125 \begin{bmatrix} 0.70 & 0.60 \\ 0.40 & 0.80 \end{bmatrix} \begin{bmatrix} 10000 \\ 20000 \end{bmatrix}$$

$$= 3.125 \begin{bmatrix} 19000 \\ 20000 \end{bmatrix}$$

$$= \begin{bmatrix} 59,375 \\ 62,500 \end{bmatrix}$$

So the industries must produce approx:

Coal output = 59,375

Steel output = 62,500

8. A fuel refiner wants to know the demand for a grade of gasoline as a function of price. The table shows daily sales y (in gallons) for three different prices. Find the least squares regression line for these data, then estimate the demand when the price is at \$3.65/gallon.

Price, x (dollars per gallon)	\$3.25	\$3.50	\$3.75
Demand, y (gallons)	4500	3750	3300